

INA219 POWER MEASUREMENT

Theory and Technical Documentation

Abstract

This project investigates the electrical power consumed by a low-voltage DC device using an INA219 current and power monitor connected to an Arduino Uno. The INA219 measures bus voltage and the voltage drop across a low-resistance shunt resistor. Using the shunt measurement and its calibration settings, the sensor determines current and provides a power reading. Measurement data are transferred to the Arduino over the I²C interface and displayed through the serial monitor. An LED-based load is used in the prototype setup. The system demonstrates a practical method for observing voltage, current, and power in a small DC circuit. The reliability of the readings depends on correct wiring, suitable sensor configuration, and operation within the limits of the hardware.

Keywords: INA219, electrical power, voltage, current, shunt resistor, Arduino Uno, I²C.

1. Introduction

Electrical power describes the rate at which a device consumes or converts electrical energy. For a DC load, power can be estimated from the voltage across the load and the current flowing through it. Since current draw can vary with a device's operating state, measuring both quantities provides a practical way to assess its electrical consumption.

This project uses an INA219 monitor and an Arduino Uno to acquire and report these electrical parameters. The sensor measures bus voltage and shunt voltage, derives current, and provides a power value. The Arduino reads the sensor over I²C and prints the measurements at regular intervals. The intended application is measurement of suitable low-voltage DC loads within the limits of the sensor breakout and circuit.

2. Objective and Scope

The primary objective is to determine the power consumed by a connected DC device by measuring its voltage and current. The project aims to measure bus voltage at the INA219 measurement point, obtain current from the shunt-voltage measurement, read the sensor's power value, and display these parameters through the Arduino serial monitor. The reported power can be checked against $P = V \times I$ when the voltage and current values refer to the same load and measurement conditions.

The prototype demonstrates periodic measurement and display. It does not regulate the supply, control the load, or calculate cumulative energy consumption.

3. Electrical Theory

3.1 Electric Charge and Current

Electric charge is a property of matter associated with electrical interactions. In a conducting circuit, mobile charge carriers move when an electric field is established. Electric current describes the rate at which charge passes a point and is measured in amperes (A). One ampere corresponds to one coulomb of charge passing a point each second.

$$I = Q / t$$

In this expression, I is current, Q is charge in coulombs, and t is elapsed time in seconds. The equation expresses average current over the stated interval; current can vary with time in a real device. The program reports current in milliamperes (mA), where 1 A = 1000 mA. Current depends on the applied voltage and the electrical characteristics of the connected load. In a series circuit, the same current flows through each

component along a single path, which is why the sensor's shunt must be placed in the load's current path to measure it.

3.2 Voltage and Potential Difference

Voltage is the difference in electric potential between two points in a circuit. It is measured in volts (V) and represents the energy transferred per unit charge as charge moves between those points. A power supply establishes a potential difference that can drive current through a connected load, provided there is a complete conducting path. An open circuit can have voltage across its terminals while carrying almost no current; current requires a complete path.

Voltage is always measured between two points. The supply voltage, the voltage at the INA219 bus input, and the voltage directly across the load may differ slightly because of voltage drops across the shunt, wires, and connectors. This distinction matters when calculating load power: the voltage used should correspond as closely as possible to the device whose consumption is being measured. A voltmeter is connected in parallel with the points being measured, whereas a current-sensing shunt is inserted in series with the load.

3.3 Resistance and Ohm's Law

Resistance describes how strongly a component opposes current and is measured in ohms (Ω). For a component that behaves approximately as a constant resistance under the conditions of measurement, Ohm's law relates voltage, current, and resistance:

$$V = I \times R$$

$$I = V / R$$

$$R = V / I$$

For a fixed voltage, a larger resistance generally results in a smaller current. This relationship is useful for analysing resistors and the INA219 shunt resistor. However, not all loads behave as fixed resistances. LEDs have a nonlinear voltage-current relationship, motors can draw different currents depending on their load and speed, and electronic devices may change their current demand as they operate. For such loads, direct measurement is often more informative than assuming a constant resistance.

Resistance also converts some electrical energy into heat. The power dissipated by a resistor can be expressed as $P = I^2R$ or $P = V^2/R$. In a practical circuit, this matters when choosing resistors and checking whether a component can safely handle the heat generated. The INA219 shunt is designed to have a low resistance so that its voltage drop and heat generation remain limited at the intended current.

3.4 Electrical Power

Electrical power is the rate at which electrical energy is transferred or converted. In a DC circuit, the power absorbed by a load can be calculated from the voltage across it and the current flowing through it:

$$P = V \times I$$

When voltage is expressed in volts and current in amperes, power is expressed in watts (W). Since the INA219 library methods used in this project return current in milliamperes and power in milliwatts, unit conversion should be considered when checking the results. Multiplying voltage in volts by current in milliamperes gives the numerical power in milliwatts. For example, a load at 5 V drawing 200 mA consumes approximately 1000 mW, equivalent to 1 W.

Power consumption can change as a device changes state or as its supply voltage changes. The readings in this project are sampled periodically, so they show the values at measurement times rather than a continuous record of every short-lived change. For a stable DC load, the sampled values can provide a useful estimate of its operating power. If voltage or current changes significantly between samples, the serial output may not capture every variation.

3.5 Power and Energy

Power and energy are related but represent different quantities. Power describes the rate of energy transfer, while energy is the total amount transferred over a period. Energy is commonly measured in joules (J) or watt-hours (Wh). If a device consumes a steady 2 W for 3 hours, its energy use is 6 Wh. One watt-hour equals 3600 joules.

Determining energy consumption from changing power readings requires recording power over time and accumulating the contribution of each interval. For a nearly constant power, energy can be estimated by multiplying power by elapsed time. The current program reports sampled power and does not perform this integration, but the existing measurements could form the basis of a future energy-monitoring extension.

4. INA219 Sensor Operation

The INA219 is a digital current-shunt and bus-voltage monitor. It combines analogue measurement circuitry, an analogue-to-digital converter (ADC), and an I²C-compatible digital interface. The sensor measures the differential voltage across a shunt resistor and the bus voltage, then makes digital measurement values available to a microcontroller. This allows the controller to retrieve electrical readings without directly measuring the small shunt voltage using a standard analogue input. The Arduino communicates with the sensor through the Wire library and the Adafruit INA219 library provides convenient functions for reading the measurements.

4.1 Shunt Resistor and Current Measurement

A shunt resistor is a low-resistance component placed in series with the load's current path. Current passing through it creates a small voltage drop. Because the resistance is known or configured for the measurement, current can be determined from this drop using Ohm's law:

$$V_{shunt} = R \times I$$
$$I = \frac{R}{V_{shunt}}$$

The shunt resistance is kept low to limit the voltage lost in the measurement path and reduce power dissipated by the resistor. The resulting voltage is small, so the sensor measures the difference between the shunt terminals with a suitable differential measurement circuit. The shunt still affects the circuit slightly and dissipates heat, with its dissipation given by $P_{shunt} = I^2 \times R_{shunt}$. A larger current produces a larger voltage and greater heat dissipation for the same resistance.

The INA219 derives current using the shunt measurement and its calibration configuration. The Adafruit library handles the conversion in the `getCurrent_mA()` method. The actual shunt resistance depends on the specific breakout board and should be confirmed from its documentation; the IC's specifications alone do not establish the safe current rating of every module. Incorrect calibration or a mismatch between the configured and actual shunt value can affect the reported current and power.

4.2 Bus Voltage

The bus-voltage channel measures voltage at the sensor's bus input relative to its ground reference. In a common high-side arrangement, the shunt is placed between the supply's positive terminal and the load, and the bus measurement point is on the load side of the shunt. The bus reading may therefore be slightly lower than the source-terminal voltage while current is flowing. The `getBusVoltage_V()` method returns this reading in volts.

4.3 Power Reading and Calibration

The INA219 provides a power reading based on its measured voltage and current information and its calibration configuration. In the software, `getPower_mW()` returns power in milliwatts. This value can be compared with $P = V \times I$ as a basic consistency check, provided that the voltage and current values correspond to the same load and relevant measurement conditions. Small differences can arise from

calibration, resolution, wiring losses, and the exact points at which voltage is measured. The result should therefore be interpreted as a measurement with finite accuracy rather than an exact value.

4.4 High-Side Measurement and Limits

High-side measurement places the shunt between the supply's positive terminal and the load. This arrangement measures current without inserting the shunt into the ground return. Correct polarity is important because the direction of current through the shunt affects the sign of the current reading. Safe operating limits depend on the INA219 IC and the complete breakout board, including its shunt resistor, PCB traces, connectors, and thermal performance. The board documentation must be checked before connecting a load; the circuit should remain within the module's specified voltage and current limits.

5. Methodology

The prototype consists of an Arduino Uno connected by USB to a computer, an INA219 module mounted on a small breadboard, and an LED-based load assembled on a separate breadboard. Jumper wires connect the sensor, controller, and load. The photograph documents this physical arrangement; exact pin-to-pin connections should be confirmed against the circuit wiring and the labels on the specific module.

The measurement procedure is as follows:

1. Assemble the Arduino Uno, INA219 module, and LED load on the breadboards.
2. Connect the INA219 to the Arduino's I²C interface using SDA and SCL, provide appropriate power, and establish a common ground.
3. Route the load current through the INA219's designated current-sensing terminals, following the module's polarity markings and wiring instructions.
4. Upload the program using the Adafruit INA219 library and initialise serial communication at 9600 baud.
5. Confirm that the sensor is detected. If initialisation fails, inspect power, ground, I²C wiring, and connections.
6. Read bus voltage, shunt voltage, current, and power; display each value with its unit in the serial monitor.
7. Repeat the measurements at approximately five-second intervals and use the readings to observe the electrical behaviour of the load.

The load current must pass through the sensor's current path for current and power measurements to be meaningful. Before powering the circuit, verify supply polarity, the board's voltage and current limits, and the controller's I²C pin mapping. This setup is intended for low-voltage DC experiments, not mains measurement.

6. Software Operation

The program starts serial communication at 9600 baud and calls `ina.begin()` to initialise the sensor. If the sensor is not detected, an error is printed and execution stops in a loop. After successful initialisation, the loop obtains four readings:

```
float shuntV = ina.getShuntVoltage_mV();
float busV = ina.getBusVoltage_V();
float current = ina.getCurrent_mA();
float power = ina.getPower_mW();
```

The returned units are millivolts (mV), volts (V), milliamperes (mA), and milliwatts (mW), respectively. The readings are printed to the serial monitor, followed by a five-second delay. Calling `ina.begin()` again at the end of every loop is generally unnecessary after successful initialisation in `setup()`.

7. Example Calculation

For a hypothetical load measured at 5.0 V and 250 mA, current is 0.25 A. The estimated power is:

$$P = 5.0 \text{ V} \times 0.25 \text{ A} = 1.25 \text{ W} = 1250 \text{ mW}$$

This is an illustrative calculation; actual values depend on the connected load and measurement conditions.

8. Accuracy and Limitations

Accuracy depends on sensor error, shunt tolerance, calibration, temperature, electrical noise, wiring, and the measurement range. The number of digits displayed does not by itself indicate measurement accuracy. Readings should be compared with an appropriate meter when validating the setup.

The sensor measures voltage at its own bus input, which may differ from the voltage directly across the load due to wiring or connector drops. A negative current reading can indicate that current flows opposite to the sensor's expected direction. Unexpected readings should prompt checks of polarity, common ground, I²C connections, the current path, and the breakout board's ratings.

9. Conclusion

The prototype demonstrates a method for measuring the electrical power consumed by a low-voltage DC load. The INA219 measures bus and shunt voltages, derives current, and provides a power reading to the Arduino Uno over I²C. The Arduino displays the results at regular intervals. Correct wiring, suitable operating limits, and awareness of calibration are necessary for reliable measurements.

References

1. Texas Instruments, *INA219 Zero-Drift, Bidirectional Current/Power Monitor With I²C Interface*, datasheet.
2. Adafruit, *INA219 Current Sensor* guide and Adafruit INA219 library documentation.
3. Arduino documentation for the Wire library, Serial communication, setup(), loop(), and delay().